



Alaska Automation Capability (AAC) Automation Requirements Document

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Focal Point

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Table of Contents

1	Introduction	1
1.1	Description.....	1
1.1.1	Purpose	1
1.1.2	Program Description.....	1
1.1.3	Program Background	2
1.1.4	Alaska Flight Service Shortfalls	2
1.1.5	Assumptions	3
1.2	Scope.....	3
2	Capability Description and Program Information	5
2.1	Operational Concept.....	5
2.2	Quantities and Location.....	5
2.3	Program Constraints and Dependencies	5
3	Functional and Performance Requirements	6
3.1	Functional Requirements	6
3.1.1	Preflight Services	6
3.1.2	Inflight Services.....	6
3.1.3	Flight Data Services/Operational Services.....	6
3.1.4	Self-Assisted Services	9
3.1.5	Electronic Flight Strip Systems	9
3.2	Alaska Data and Information Requirements	14
3.2.1	General Alaska Data Requirements.....	15
3.2.2	Alaska FSS Local Area Knowledge requirements	15
3.3	Performance Requirements.....	16
3.3.1	System Performance.....	16
3.3.2	Performance Management.....	24
3.3.3	Recovery Requirements	24
3.4	Reliability, Maintainability, Availability, and Operational Continuity	24
3.4.1	Reliability.....	25
3.4.2	Maintainability.....	25
3.4.3	Availability	25
3.4.4	Continuity of Operations	25
4	Integration Requirements.....	26

4.1	Physical Integration.....	26
4.1.1	Environmental Requirements	26
4.1.2	Land and Facility	26
4.1.3	Telecommunications.....	26
4.2	Functional Integration.....	26
4.2.1	Interfaces to Other FAA Enterprise Architecture Elements.....	26
4.2.2	External Interfaces	27
4.2.3	Spectrum Requirements	27
4.3	Human Factors	28
4.3.1	Usability	28
4.3.2	Human-System Interfaces	28
4.3.3	Computer-Human Interfaces (CHI)	28
4.3.4	General Control and Visual Indicator Information.....	28
4.4	Employee Safety and Health	28
5	Security and Safety Requirements.....	29
5.1	Facility Security.....	29
5.2	Personnel Security	29
5.3	Information Systems Security.....	29
5.3.1	Security Management.....	29
5.3.2	Security Assessments.....	29
5.3.3	Security-Related System Support	29
5.3.4	Security and Information Access.....	30
5.3.5	System and Information Monitoring.....	30
5.3.6	System and Information Protection.....	30
5.3.7	Privacy Information Protection	31
5.4	System Safety.....	31
5.4.1	Existing Controls.....	31
5.4.2	Safety Requirements	31
6	Quality Management Requirements.....	32
6.1	Quality Assurance.....	32
6.2	Configuration Management	32
6.2.1	Configuration Management Program	32
6.2.2	Configuration Management Planning.....	32
7	Test and Evaluation Requirements	33

7.1	Planning and Test Design Requirements.....	33
7.2	Developmental Test (DT) Requirements	33
7.3	Operational Test Requirements.....	34
7.3.1	Interoperability Testing Requirements.....	33
7.3.2	Operational Suitability Requirements.....	34
7.4	Site Test Requirements	34
7.5	Reporting Requirements.....	34
7.6	Test Support Requirements	35
7.7	Critical Operational Issues.....	35
8	In-Service Support and Management.....	36
8.1	Integrated Logistics Support	36
8.2	Maintenance Planning.....	36
8.3	Packaging, Handling, Storage, and Transportation.....	36
8.4	Support Equipment Maintenance	36
8.4.1	Tools and Test Equipment	36
8.4.2	Maintenance Operations	36
8.5	Training, Training Support, and Personnel Skills.....	36
8.5.1	FAA Academy Training System.....	36
8.5.2	AAC Training, Training Support, and Personnel Skills	36
8.6	Technical Data	39
Appendix A	Critical Performance Requirements	40
Appendix B	Acronyms and Glossary.....	41
B.1	Acronyms and Abbreviations.....	41
B.2	Terms and Definitions	48
Appendix C	Reference Documents	54
C.1	FAA Documents	54
C.2	Other Government Documents.....	55
C.3	Non-Government Documents	55

Index of Figures

No table of figures entries found.

Index of Tables

Table 2 - Product Acquisition Performance Requirements.....	16
Table 3 - FSS Site Peak Loading.....	16
Table 4 - FSS Site Peak Concurrent Transaction Rates	17
Table 5 - Operational Performance Parameters	18
Table 6 - Critical Performance Requirements.....	40
Table 7: Acronyms and Abbreviations.....	41
Table 8: Terms and Definitions	48

1 Introduction

1.1 Description

1.1.1 Purpose

This document presents the functional requirements for the Alaska Automation Capability (AAC), identifying essential system functional capabilities and performance measures necessary to support flight services in Alaska.

This Automation Requirements Document is provided to establish operational context and baseline insight into the Government's current understanding of capability needs. For purposes of this Challenge-Based Acquisition phase, respondents are not required to demonstrate full compliance with each requirement. Rather, respondents should use this material to inform their conceptual approach and describe how their proposed solution would address the Government's objectives at a capability level.

Binding performance obligations, final requirement refinement, and compliance validation will be defined as appropriate, in any subsequent production award instrument.

1.1.2 Program Description

Flight Service collects, processes, interprets, translates, and disseminates aeronautical and meteorological information to promote safe and expeditious flights through pilot weather briefings, inflight weather advisories, and flight plan filing. Flight Service relays Instrument Flight Rules (IFR) and Special Visual Flight Rules (SVFR) clearances as appropriate and provides applicable information on preferred IFR routes as well as departure and arrival procedures. In addition, Flight Service provides Search and Rescue (SAR) services integrated with commercial Global Positioning Service (GPS) aircraft tracking, and orientation services to Visual Flight Rules (VFR) flights. Flight Service is an important source of information flow to the National Airspace System (NAS) through the filing of flight plans, aircraft movement messages, and the collection and dissemination of Notices to Air Missions (NOTAMs), and Pilot Weather Reports (PIREPs).

The General Aviation (GA) community represents the primary customer base and generates the majority of all service requests. These include requests for preflight planning, pilot briefings, flight plan filing, meteorological and aeronautical information, and inflight orientation services. Flight Service supports disaster relief, medical evacuations, weather and traffic reporting, and business/personal travel to thousands of airports/landing areas throughout the NAS. Military, law enforcement, commercial, corporate, air taxi, U.S. Customs and Border Protection (CBP), Civil Air Patrol, Medevac and others also benefit from services provided by Flight Service.

Flight Service also supports other FAA Air Traffic Organization (ATO) service units. Flight Service Specialists (Specialists) and Certified Professional Controllers (CPCs)¹ relay communications from pilots to Terminal and En Route controllers (when no direct means exist)

¹ Flight Service Specialists provide contracted flight services for the contiguous United States (CONUS) through an existing contract, while Certified Professional Controllers provide flight services in Alaska as FAA FTEs.

and assist in data collection and data entry. Flight Service coordinates with Technical Operations regarding equipment (e.g., radio, NAVAIDs) status information. Flight Service has historically provided meteorological data, aeronautical information, and flight planning services to its pilot customers in two ways. The traditional service delivery (human-assisted) consists of pilots contacting a Specialist by phone or radio. Specialists access a variety of meteorological products, aeronautical data, and other related information based on the pilot's requested route and share that information with the pilot orally. Alternatively, pilots can obtain preflight information and file flight plans through an online web portal. Pilots accessing preflight information online are responsible for obtaining and understanding all available information about their flight prior to filing a flight plan.

Over the past decade, emerging technologies in communications, personal computing, and mobile devices have provided opportunities to deliver flight services more efficiently to users. User preferences and demands are reflected in the continuing decline in the use of human-assisted services as pilots take advantage of automated and enhanced technologies (e.g., integrated weather products, electronic flight bags, Automatic Dependent Surveillance – Broadcast [ADS-B]) available through the FAA and private sector.

1.1.3 Program Background

In 2005, the FAA awarded a contract to a commercial provider (Lockheed Martin/Leidos) for preflight planning support and in-flight advisory assistance in the contiguous United States (CONUS), Hawaii, and Puerto Rico. That contract, known as Flight Service for the 21st Century (FS21), realized operational improvements (OIs) along with economies in staffing and infrastructure.

In Alaska, flight services are provided by federal employees using the Operational and Supportability Implementation System (OASIS II). The OASIS II system consists of a leased automation system operated by Federal CPCs providing the full range of flight services in Alaska.

1.1.4 Alaska Flight Service Operational Shortfalls

Alaska Flight Service (AFS) relies on voice switch and automation systems, both of which are facing current operational and technical gaps and face increasing risks that could fail, preventing Flight Service from performing the critical services in Alaska. The legacy automation system, OASIS II, provides the capability to acquire, store, retrieve and display information to compile and deliver the services through voice and data connectivity with pilots and FAA ATC facilities. However, the system has been maintained but not modernized or improved for many years and is using aging hardware that have reached end-of-life. The OASIS II managed services contract is currently extended with annual and semi-annual option years through 2028.

The legacy automation system faces several shortfalls including system limitations such as data management, communication, and bandwidth limitations which lead to labor and management inefficiencies and minor safety impacts. Additionally, these limitations contribute to administrative inefficiencies which lead to costs increases in administrative affairs and staffing resource management. This Automation Requirements Document is intended to address key operational shortfalls by implementing a modernized automation solution that deploys updated technology and supports the FAA's objectives to greatly improve Flight Service capabilities.

1.1.5 Assumptions

The following assumptions apply to the AAC acquisition:

- The solution will be a rapidly deployable, web-based cloud hosted Flight Service system.
- The FAA's NAS Trusted Cloud (NTC) will be in place and accepting client services in time for this acquisition.
- The FAA will modernize its communications infrastructure to utilize IP-based network technologies. This will include terrestrial-based communications to Alaska Flight Service Stations (FSS).²
- FAA Telecommunications Services will be transitioned in Alaska from FAA Telecommunications Infrastructure (FTI) to FAA Enterprise Network Services (FENS) prior to AAC site implementation activities.
- Demand for services in Alaska will continue at or above current levels.
- The Flight Service delivery model for Alaska will remain unchanged.
- Flight Service Certified Professional Controller (CPC) training will continue to be conducted at the AFS Training Academy in Kenai.
- Maintenance training can be conducted at a vendor facility or FAA training facility.
- FAA Technical Operations will provide first-level system support for all hardware equipment contained in Alaska facilities.
- All NOTAM management activities will be supported through the FAA's cloud-based NOTAM Management Service (NMS).
- Self-assisted services for Alaska will be provided by the existing FS21 system.
- The Government will provide all telecommunication services and access necessary for the performance of AAC.

1.2 Scope

The scope of AAC and these requirements includes providing flight services for Alaska. The flight services should be distinct from pilot-facing subscription products, even if derived from those products. Flight Services include:

- Providing preflight planning services, which includes telephone based, preflight pilot briefings that accompany flight planning and filing both IFR and VFR flight plans via phone calls or through self-assisted means.
- Managing existing IFR/VFR flights plans, including active VFR tracking, overdue alerts, and flight plan close-outs.
- Maintaining and providing continual weather broadcasts on selected Navigation Aids (NAVAIDs).
- Integrating FAA authoritative weather, aeronautical data, and flight plan services as outlined in FAA Order 7110.10.
- Issuing and cancelling Notice to Airmen (NOTAM).

² All Alaska FSS are expected to receive fiber-optic networking infrastructure by contract implementation, apart from the McGrath FSS.

- Relaying clearances and providing Airport Advisory services at non-towered or part-time towered airports.
- Developing, translating, processing, and coordinating aeronautical, meteorological and aviation information, to advise, notify and educate customers.
- Supporting safety operations, such as emergency services, Search and Rescue, and pilot orientation services.

2 Capability Description and Program Information

2.1 Operational Concept

The Respondent-provided web-based, cloud automation system will incorporate functionality that supports the FAA CPC delivery of flight services for any Alaska Flight Plan Area (FPA), also known as an Area of Responsibility (AOR), from any workstation at an Alaska FSS facility. Provision of these services will comply with FAA Order JO 7110.10, Flight Service. The automation will combine capabilities currently supporting Flight Service operations including self-assisted (e.g., web enabled) services along with legacy functions necessary for the conduct of operations in Alaska. Support will be provided to Alaska Flight Service CPCs and supervisors for applications such as weather briefings, inflight services, flight data services, quality assurance and control (QA/QC) processes, and supervisory and administrative functions.

2.2 Quantities and Location

Flight Service operations in Alaska are currently supported by 17 facilities — three Hub FSSs and 14 Satellite FSSs. Hub FSSs include Kenai, Fairbanks, and Juneau. Satellite FSSs include Barrow, Deadhorse, Kotzebue, Nome, Northway, Iliamna, Talkeetna, McGrath, Dillingham, Cold Bay, Homer, Palmer, Sitka, and Ketchikan. At night and during seasonal closings, Hub FSSs subsume Satellite FSS operations. When the Satellite FSSs close within a flight plan area, the radio frequencies and telephone lines at the Satellite FSSs are transferred to the assigned Hub FSS.

Each facility has several different types of working positions, including In-Flight, Pre-Flight, Flight Data, Supervisor, and Maintenance positions. Additionally, some facilities currently have training positions and spare working positions. To avoid redundancy, the current listing of Alaska FSSs and number of positions is provided in other program documentation.

2.3 Program Constraints and Dependencies

Operationally, the AAC web-based, cloud automation system requires functionality that ensures compliance with the current FAA Order JO 7110.10. This order, which is periodically revised, prescribes the procedures and phraseology for use by Flight Service personnel providing services to the users of the NAS. The Program Office for AAC will maintain awareness of updates to this order to ensure the AAC functional capabilities provided by the Respondent remain in alignment with the order.

3 Functional and Performance Requirements

Each requirement (“must” statement) in this document provides a permanent identifying tag in the form of “AAC-####” where the #### indicates a 4-digit value that is unique to the requirement and is persistent through the life of the requirement. This is used for configuration management and requirement life-cycle control.

3.1 Functional Requirements

Flight services are categorized into the following areas: Preflight Services, Inflight Services, Flight Data/Operational Services, and Self-Assisted Services.

3.1.1 Preflight Services

The Respondent must [AAC-1004] provide data for Alaska CPCs to provide Standard, Abbreviated, Outlook, and Delta Weather Briefings for assigned AORs in accordance with FAA Order 7110.10.

The Respondent must [AAC-1006] provide the capability for Alaska CPCs to perform flight planning functions in accordance with FAA Order 7110.10, Flight Services.

The Respondent must [AAC-1008] provide the capability for Alaska CPCs to perform flight planning functions in accordance with NAS-IR-23040001.

The Respondent must [AAC-1010] provide the capability for Alaska CPCs to perform flight planning functions in accordance with ICAO Rules of the Air and Air Traffic Services document 4444.

3.1.2 Inflight Services

The Respondent must [AAC-1012] provide the capability for Alaska CPCs to perform Inflight functions in accordance with FAA Order 7110.10.

3.1.3 Flight Data Services/Operational Services

The Respondent must [AAC-1014] provide functionality and data necessary for Alaska CPCs to perform flight data services in accordance with FAA Order JO 7110.10, Flight Services.

3.1.3.1 National Security

The Respondent must [AAC-0084] support aviation activities pertaining to national security in accordance with current national security restrictions and guidance.

The Respondent must [AAC-0086] protect national security and sensitive information in accordance with FAA Order 1600.2, Classified National Security Information.

The Respondent must [AAC-0087] release national security and sensitive information to authorized parties as following:

- As directed by the Government
- As addressed in FAA Order 1600.2, Classified National Security Information
- As addressed in FAA Order 1600.69, FAA Facility Security Management Program

3.1.3.2 Search and Rescue (SAR)

The Respondent must [AAC-4132] provide the capability for Alaska CPCs to perform Search and Rescue (SAR) functions in accordance with FAA Order 7110.10.

The Respondent must [AAC-4134] allow Alaska CPCs to manage SAR messages received from the Enhanced Special Reporting Service (eSRS).

The Respondent must [AAC-1460] accept SAR messages from external facilities.

The Respondent must [AAC-1481] automatically address SAR messages for transmission to appropriate facilities, appending address(es) into the appropriate field based upon the type of SAR message.

The Respondent must [AAC-4136] make available for edit improperly formatted SAR messages.

3.1.3.3 Data Recording and Retention

The Respondent must [AAC-0092] log all user actions performed on the automation system.

The Respondent must [AAC-0093] log all user actions performed on the web portal/application.

The Respondent must [AAC-0094] record all communications from operational positions.

The Respondent must [AAC-0096] maintain all system logs in accordance with FAA Order JO 7210.3, Facility Operation and Administration.

The Respondent must [AAC-0097] maintain all system recordings in accordance with FAA Order JO 7210.3, Facility Operation and Administration.

3.1.3.4 System Configuration

3.1.3.4.1 Work Position Configurations

The Respondent must [AAC-2230] allow authorized FAA personnel the ability to create work position configurations which support Alaska FS operations based on Area of Responsibility, Work Position, and Flight Service Facility.

The Respondent must [AAC-2232] permit the definition of work position configurations in the configuration database sufficient to support Alaska flight service operations.

Note: Alaska flight service currently supports 11 functional positions for CPCs and supervisors (Administrator, Flight Data, Inflight Controller in Charge (CIC), Inflight, Maintenance, Operational Supervisor (OS) CIC, Preflight, Quality Assurance/Quality Reliability Officer (QRO), Test, Training, Other) and is operational at 17 facilities.

The Respondent must [AAC-2240] recover from a system restart with the future Alaska FS workstation configurations available for specialist initiation at each position.

3.1.3.4.2 Supervisor Work Assignments

The Respondent must [AAC-2254] permit any Alaska FS workstation to be used to access supervisory functions.

The Respondent must [AAC-2256] allow a FAA FS Supervisor the ability to assign Alaska configurations to staff to support daily work assignments based on Specialist: AOR qualifications, work schedule, facility, and/or Work Position.

3.1.3.4.3 Specialist Work Assignments

The Respondent must [AAC-2272] allow FAA Specialist the ability to configure the Controller Work Position based on their shift assignment.

The Respondent must [AAC-2274] display the current user identification and their currently assigned FS role and AOR.

The Respondent must [AAC-2276] allow FAA Specialist to configure the CWP to support user preferences.

The Respondent must [AAC-2278] allow the FAA specialist to store user preferences.

3.1.3.4.4 System Reconfiguration

The Respondent must [AAC-2286] prevent the interruption of service during reconfiguration on those positions and external interfaces not affected by the reconfiguration.

The Respondent must [AAC-2288] ensure that reconfiguration process does not disconnect any established call at the position until reconfiguration is accepted.

The Respondent must [AAC-2290] require no more than one minute to prepare reconfiguration data, measured from the time reconfiguration is invoked by the operator to the time that the System requests the first position acknowledgment of reconfiguration, or to the time that the System performs the reconfiguration, for resources that do not require acknowledgment.

3.1.3.5 Monitor and Control

The Respondent must [AAC-2325] provide FAA Airway Transportation System Specialist (ATSS) personnel monitoring and control capabilities for all AAC equipment located in FAA facilities.

3.1.3.5.1 System Monitor and Control

The Respondent must [AAC-2326] ensure that the initial state of each installed device on startup is configurable as being online, offline, or maintenance busy.

The AAC Monitor and Control (M&C) function must [AAC-2210] monitor system health.

The AAC M&C function must [AAC-2211] monitor system performance.

3.1.3.5.2 Supervisor Monitor

The Respondent must [AAC-2224] display the status of any ongoing reconfigurations on the authorized access levels' workstation.

3.1.3.6 Fault Detection and Isolation

The Respondent must [AAC-2304] perform fault and failure detection and isolation.

The Respondent must [AAC-2306] ensure that fault detection does not interfere with system operations.

The Respondent must [AAC-2308] report all detectable faults to the maintenance workstation(s) and to the maintenance log simultaneously.

The Respondent must [AAC-2312] permit the initiation of manual diagnostic testing.

3.1.3.7 Maintenance Logging

The Respondent must [AAC-2316] generate a maintenance log that contains faults detected by the system fault detection features.

The AAC maintenance log must [AAC-2318] contain restoration activities.

The AAC maintenance log must [AAC-2320] contain reconfiguration events.

The AAC maintenance log must [AAC-2322] be capable of storing at least 90 days of maintenance history.

The Respondent must [AAC-2324] permit authorized users to print the maintenance logs from any Alaska FSS.

3.1.4 Self-Assisted Services

Self-Assisted Services in Alaska will be provided by the existing FS21 system.

3.1.5 Electronic Flight Strip Systems

The AAC will provide Alaska CPCs an electronic input device and output display to create and edit digital flight progress strips. The CPCs will also be able to create digital flight progress strips from flight plan masks in an Active Flight Workspace bay, and in Inflight Work bays. Digital flight progress strips will allow the CPCs to perform designated flight planning and flight service support functions directly from the electronic input device. These functions include, but are not limited to, filing, logging, weather briefing, canceling, closing, amending, and activating a flight plan; and initiating search and rescue operations.

3.1.5.1 Digital Flight Progress Strips

The AAC Electronic Flight Strips system (EFS) must [AAC-1318] provide an input device for managing digital flight progress strips.

Note: In the above requirement, the actions associated with “managing” digital flight progress strips include, but are not necessarily limited to creating, editing, deleting, copying, and manipulating.

The AAC EFS must [AAC-1319] provide the capability to create a digital flight progress strip using the electronic input device.

The digital flight progress strip must [AAC-1320] include an Aircraft ID field.

The digital flight progress strip must [AAC-1321] include an Aircraft Type/Special Equipment field.

The digital flight progress strip must [AAC-1322] include a TAS/Alt field.

The TAS/Alt field must [AAC-1323] include an Airspeed field.

The TAS/Alt field must [AAC-1324] include an Altitude field.

- The digital flight progress strip must [AAC-1325] include a Departure field.
- The digital flight progress strip must [AAC-1326] include a Route field.
- The digital flight progress strip must [AAC-1327] include a Destination field.
- The digital flight progress strip must [AAC-1328] include a Time of Departure field.
- The digital flight progress strip must [AAC-1329] include an ETA field.
- The digital flight progress strip must [AAC-1330] include an ETE field.
- The digital flight progress strip must [AAC-1331] include a Flight Type field.
- The digital flight progress strip must [AAC-1332] include a Remarks field.
- The digital flight progress strip must [AAC-1333] provide the capability to record a Time of Contact with a Pilot when communication with a Pilot is established based on user input.
- The digital flight progress strip must [AAC-1334] include an Information Received from Pilot/Another Facility field.
- The digital flight progress strip must [AAC-1335] include a Data Issued to the Aircraft field.
- The digital flight progress strip must [AAC-1336] provide a capability to record a Number on Board for a flight based on user input.
- The digital flight progress strip must [AAC-1337] provide a capability to record current Flight Rules for a flight based on user input.
- The digital flight progress strip must [AAC-1338] provide a capability to record Fuel on Board for a flight based on user input.
- The digital flight progress strip must [AAC-1339] provide a capability to record Pilot Data based on user input.
- The digital flight progress strip must [AAC-1340] provide a capability to record Responsible FSS Data.
- The digital flight progress strip must [AAC-1341] provide a capability to record Notification Data.
- The digital flight progress strip must [AAC-1342] include a Comm/Nav Equipment field.
- The digital flight progress strip must [AAC-1343] include a Transponder field.
- The digital flight progress strip must [AAC-1344] include a Wake Turbulence field.
- The AAC EFS must [AAC-1345] provide the capability to display digital flight progress strip records.
- The AAC EFS must [AAC-1346] provide the capability to edit digital flight progress strip records based on user input.
- The AAC EFS must [AAC-1347] provide the capability to perform flight planning functions based on user input.

The AAC EFS [AAC-1348] must provide the capability to file flight plans based on the digital flight progress strips records and data for a flight. The handwriting input device must [AAC-1349] log the following:

- The log time must [AAC-1350] default to the current time.

- The AAC EFS must [AAC-1351] log airport advisories.

- The AAC EFS must [AAC-1352] log pilot briefings.

- The AAC EFS must [AAC-1353] log Special Visual Flight Rules (SVFR) clearances issued.

- The AAC EFS must [AAC-1354] log IFR clearances issued.

- The AAC EFS must [AAC-1355] log aircraft contacts.

- The AAC EFS must [AAC-1356] log radio contacts.

- The AAC EFS must [AAC-1357] log Request/Response information.

The handwriting input device must [AAC-1358] provide the capability to request a weather briefing.

The AAC EFS must [AAC-1359] cancel a flight plan based on user input.

The AAC EFS must [AAC-1360] close a flight plan based on user input.

The AAC EFS must [AAC-1361] provide the capability to amend a flight plan based on user input.

The AAC EFS must [AAC-1362] provide the capability to activate a flight plan based on user input.

The AAC EFS must [AAC-1363] provide the capability to search for a digital flight progress strip using the handwriting input device.

The AAC EFS must [AAC-1364] provide the capability to create a digital flight progress strip template.

- The digital flight progress strip template must [AAC-1365] include a Flight Type field.

- The digital flight progress strip template must [AAC-1366] include a Flight Rules field.

- The digital flight progress strip template must [AAC-1367] include an Aircraft ID field.

- The digital flight progress strip template must [AAC-1368] include an Aircraft Type field.

- The digital flight progress strip template must [AAC-1369] include an Airspeed field.

- The digital flight progress strip template must [AAC-1370] include an Altitude field.

- The digital flight progress strip template must [AAC-1371] include a Departure field.

- The digital flight progress strip template must [AAC-1372] include a Route field.

- The digital flight progress strip template must [AAC-1373] include a Destination field.

The digital flight progress strip template must [AAC-1374] include a Time of Departure field.

The digital flight progress strip template must [AAC-1375] include an ETA field.

The digital flight progress strip template must [AAC-1376] include a ETE field.

The digital flight progress strip template must [AAC-1377] include a Number on Board field.

The digital flight progress strip template must [AAC-1378] include a Fuel Capacity field.

The digital flight progress strip template must [AAC-1379] include a Comm/Nav Equipment field.

The digital flight progress strip template must [AAC-1380] include a Transponder field.

The digital flight progress strip template must [AAC-1381] include a Pilot Data field.

The digital flight progress strip template must [AAC-1382] include a Wake Turbulence field.

The AAC EFS must [AAC-1383] provide the capability to store digital flight progress strip templates by:

- a. Site
- b. User

The AAC EFS must [AAC-1386] provide the capability to edit digital flight progress strip templates.

The AAC EFS must [AAC-1387] provide the capability to delete digital flight progress strip templates.

The AAC EFS must [AAC-1388] provide the capability to display digital flight progress strip templates by:

- a. Site
- b. User

The AAC EFS must [AAC-1391] provide the capability to copy Site digital flight progress strip templates to User digital flight progress strip templates.

The AAC EFS must [AAC-1392] provide the capability to copy User digital flight progress strip templates to Site digital flight progress strip templates.

The AAC EFS must [AAC-1393] provide the capability to display and copy another User's digital flight progress strip templates.

The AAC EFS must [AAC-1394] provide the capability to copy another User's digital flight progress strip templates to the User Storage of the logged in User.

The AAC EFS must [AAC-1395] provide the capability to copy another User's digital flight progress strip templates to the Facility Storage of the logged in User.

The AAC EFS must [AAC-1396] provide the capability to display another User's digital flight progress strip RID data.

The AAC EFS must [AAC-1396.5] provide the capability to copy another User's digital flight progress strip RID data.

3.1.5.2 Electronic Strip-Bay

The AAC EFS will provide CPCs a configurable electronic strip-bay, like physical flight progress strips and bay. The digital flight progress strip display will allow the CPCs to configure flight strip layouts into partitions corresponding to separate airports or other user-defined groupings.

The AAC EFS must [AAC-1422] provide a user configurable electronic strip-bay layout containing the following digital flight progress strips:

- a. Proposed
- b. Inbound
- c. Overflight
- d. Separator

The AAC EFS must [AAC-1427] provide the capability to divide the electronic strip-bay display into the following sections:

- a. Proposed List
- b. Inbound List
- c. Active Flight Workspace
- d. Other Active Flight Workspaces
- e. Common Area

The AAC EFS must [AAC-1433] display digital flight progress strips at the top of the electronic strip-bay when added.

The AAC EFS must [AAC-1434] visually indicate when digital flight progress strips are added to the electronic strip-bay.

The AAC EFS must [AAC-1435] visually indicate if the flight plan is on the Inbound List.

The AAC EFS must [AAC-1436] visually indicate if the flight plan is on the Proposed List.

The AAC EFS must [AAC-1437] provide the capability to display individual digital flight progress strips in the active flight workspace.

The AAC EFS must [AAC-1438] provide the capability to minimize an individual digital flight progress strip in the active flight workspace.

The AAC EFS must [AAC-1439] provide the capability to display abbreviated digital flight progress strips.

The AAC EFS must [AAC-1440] provide the capability to store digital flight progress strips in the electronic strip-bay.

The AAC EFS must [AAC-1441] provide the capability to delete a digital flight progress strip from the electronic strip-bay.

The AAC EFS must [AAC-1442] provide the capability to manually arrange the digital flight progress strips in the electronic strip-bay.

The AAC EFS must [AAC-1443] provide the capability to automatically save the contents of the electronic strip-bay upon logoff.

The AAC EFS must [AAC-1444] provide the capability to automatically retrieve the information from the electronic strip-bay upon logon.

3.1.5.3 Active Flight Workspace

The active flight workspace will allow the Alaska CPC to create digital flight progress strips without using the handwriting input device. It also will allow the CPC to edit existing digital flight progress strips without using the handwriting input device.

The AAC EFS must [AAC-1445] include an Active Flight Workspace associated with the electronic strip-bay.

The AAC EFS must [AAC-1446] populate the Active Flight Workspace with the current active digital flight progress strip.

The AAC EFS must [AAC-1447] provide the capability to create digital flight progress strips in the Active Flight Workspace.

The AAC EFS must [AAC-1448] provide the capability to edit digital flight progress strips in the Active Flight Workspace.

3.1.5.4 Electronic Flight Strip Configuration/Adaptation

The AAC EFS must [AAC-1364] provide the capability to create a digital flight progress strip template.

The AAC EFS must [AAC-1391] provide the capability to copy Site digital flight progress strip templates to User digital flight progress strip templates.

The AAC EFS must [AAC-1392] provide the capability to copy User digital flight progress strip templates to Site digital flight progress strip templates.

The AAC EFS must [AAC-1393] provide the capability to display and copy another User's digital flight progress strip templates.

The AAC EFS must [AAC-1394] provide the capability to copy another User's digital flight progress strip templates to the User Storage of the logged in User.

The AAC EFS must [AAC-1395] provide the capability to copy another User's digital flight progress strip templates to the Facility Storage of the logged in User.

3.2 Alaska Data and Information Requirements

To support Alaska CPC provided flight services, AAC will maintain aeronautical and weather dynamic data, maintain a database of prestored flight plans, maintain alert queues for

improperly formatted messages and safety related NOTAMs received, edits of flight data, updates flight plan information, and maintain a NOTAM database.

3.2.1 General Alaska Data Requirements

The Respondent must [AAC-1544] maintain static reference data as required by Order JO 7210.3.

The Respondent must [AAC-1546] maintain prestored flight plans which will support Alaska CPC provided services.

The Respondent must [AAC-1547] maintain frequently used pilot/aircraft owner information which will support Alaska CPC provided services.

The Respondent must [AAC-1548] store pilot/aircraft owner contact information using AES-128 encryption.

The Respondent must [AAC-1550] maintain frequently used information pertinent to an aircraft for use for with automatically filling common flight plan fields and for search and rescue procedures.

The Respondent must [AAC-1763] maintain an Alaska CPC Inflight Contact Database.

The Respondent must [AAC-1767] maintain an Alaska CPC Traffic Count Database.

The Respondent must [AAC-1687] provide two an integrated printers at each Alaska Hub FSS and one at each Alaska Satellite FSS.

3.2.2 Alaska FSS Local Area Knowledge requirements

The Respondent must [AAC-2127] provide the capability to store local knowledge associated with geographic locations in the form of static data.

The Respondent must [AAC-2128] archive a copy of the local knowledge upon creation.

The Respondent must [AAC-2130] provide the capability to display local knowledge associated with geographic locations by any site.

The Respondent must [AAC-2131] overlay local knowledge onto any earth locatable graphic map.

The Respondent must [AAC-2136] provide the capability to edit local knowledge associated with geographic locations by any site.

The System must [AAC-2137] archive a copy of the local knowledge upon editing.

The Respondent must [AAC-2139] provide the capability to delete local knowledge associated with geographic locations by any site.

The Respondent must [AAC-2140] retain the archived copies upon deletion.

The Respondent must [AAC-2141] provide the capability to copy local knowledge overlay label text from a list of nearby local knowledge overlays.

3.3 Performance Requirements

This section identifies the performance requirements, the definition of the measurements, and a definition of the loading conditions that apply to each performance requirement and measurement.

3.3.1 System Performance

The AAC system must [AAC-3307] meet, as a minimum, the performance specifications as provided in Table 1, Table 2, Table 3, and Table 4 -

Table 1 - Product Acquisition Performance Requirements

Products	Acquisition Rates	Expected Product Rate and Frequency at FSS Site
WSR-88D products	Products are acquired at the rate and frequency they are made available from the WSR-88D.	Product updates every 5,6, or 10 minutes based on individual radar site mode.
Satellite products	Satellite image data is acquired at the rate and frequency they are made available from the satellite.	1 product update every 15 minutes
Gridded products	Products are acquired at the rate and frequency they are made available from the source.	Gridded data updated every 12 hours
Graphic products	Products are acquired at the rate and frequency they are made available from the source.	Product updates every 3 hours to 12 hours based on individual product.
A/N weather products	Products are acquired at the rate and frequency they are made available from the source.	Product update as defined in NAS-IR-23040001.

The System will disseminate the products received from the source to all AFSSs within 30 seconds on average from the receipt of the last byte of data.

Functional performance requirements are defined for the various processing that occurs at the FSS site. Loading is based on a site with between 10 and 30 workstations, A/N weather data being received and processed, and weather graphic data being received and products generated.

Table 2 - FSS Site Peak Loading

Item	Size (bytes)	Average Number Per Peak Hour	Total Size/Hr (bytes)
Route Oriented Briefing	5000	205	1.02 Mbytes
Pilot Weather Briefing	1500	107	160 Kbytes
Flight Plans	1000	150	150 Kbytes

Item	Size (bytes)	Average Number Per Peak Hour	Total Size/Hr (bytes)
Aircraft Contacts	40	94	3760 bytes
Miscellaneous	100	1190	119 Kbytes

Column 1 of Table 2 (Item) identifies the items or types of transactions the Automation System is expected to process. Column 2 (Average Number Per Peak Hour) represents the average number of transactions to be processed by the System for the Peak Hour by an FSS. During peak operations, certain types of transactions identified in Table 2 may be imposed on the system concurrently. These concurrent transactions imposed on the system may be either manually input by the specialist or prescheduled by the specialist to occur at the same time. In either case, the System will be capable of receiving up to 58 concurrent transactions. Due to receipt of these types of concurrent transactions, the System will be capable of scheduling concurrently up to 174 messages for transmission.

For transmission, concurrent transactions are defined as the number of messages with the same scheduled transmit time. For reception, concurrent transactions are defined as the number of simultaneous incoming messages. As a minimum, these concurrent transactions should be completed two minutes prior to the scheduled retransmit time.

Some of the transactions in Column 1 of Table 2, when processed by the System, will cause the system to generate external messages to remote system interfaces. As stated above, the System will be capable of scheduling concurrently up to 174 messages for transmission. Table 3 further defines the mix of concurrent transactions and their schedule for transmission.

Table 3 - FSS Site Peak Concurrent Transaction Rates

# of Concurrent Message Types	# of External Messages Scheduled for Transmission
10 Prestored Flight Plans to a single destination address, one closure point address, along with a copy to AMOC	30
Greater than or equal to 40 Flight Plans concurrently scheduled in the system using the IFR or FSS Transmit times to a single destination address, with a copy to AMOC and a copy to one Closure Point Address	Greater than or equal to 120
Greater than or equal to 8 General Facility messages of 15 lines (Miscellaneous) to a single destination address, one closure point address, along with a copy to AMOC	Greater than or equal to 24

Table 4 - Operational Performance Parameters

Requirement Description	Condition(s) (Product Sizes May Exceed Specified Values)	Time (seconds)		
		Mean	95 th %-tile	99 th %-tile
a) Receive, Process and Store Products	From receipt of last byte of data to completion of product storage			
a.1) Single Non-breakdown A/N Weather Product	500 byte product	1	1.8	2.5
a.5) Graphic Product	All non-imagery graphic products	3	5.6	7.5
a.6) Individual Radar Product		3	5.6	7.5
a.7) National Radar Mosaic Product		7	10.5	14
a.8) GOES Satellite Imagery		3	5.2	7.5
a.9) Polar Orbit Satellite Imagery		3	5.2	7.5
b) A/N Weather and NOTAM Product Display	From data request entry on the user workstation to beginning of product display			
b.1) Weather Product	Single 500 byte product	2	3.7	5
b.2) NOTAM Product	Single 500 byte product	2	3.7	5
b.3) Standard Route Weather Briefing	500nm route, 2 legs, 100nm route corridor width, 400nm winds aloft corridor width, all required briefing items	2.5	4.6	6.2
b.4) Standard Area Weather Briefing Products	Area radius 300nm, low altitude, all required briefing items	2.5	3.7	5
b.5) Outlook Route Weather Briefing Products	500nm route, 2 legs, route corridor width 100nm, all required briefing items, except no current data	2.5	4.6	6.2
b.6) Outlook Area Weather Briefing Products	Area radius 300nm, low altitude, all required briefing items	2.5	4.6	6.2

Requirement Description	Condition(s) (Product Sizes May Exceed Specified Values)	Time (seconds)		
		Mean	95 th %-tile	99 th %-tile
b.7) Product Weather Briefing (By LOCID)	10 LOCIDs, 3 product types	2	3.7	5
b.8) Product Weather Briefing (By State)	1 state, 1 product type, minimum of 50 products	2	3.7	5
b.9) Product Weather Briefing (Route)	500nm route, 2 legs, route corridor width 100nm, 2 product types	2.5	4.6	6.2
b.10) Product Weather Briefing (Area)	Area radius 300nm, 2 product types	2.5	4.6	6.2
b.11) Product Directory	Single product type	1	1.8	2.5
c) Graphic and Imagery Product Display	From data request entry on the user workstation to beginning of display			
c.1) Graphic Weather Product	All non-imagery graphic products, includes map background	2	3.7	5
c.2) Graphic Weather Product	All non-imagery graphic products, includes map background and route/area overlay.	2	3.7	5
c.3) Automatically Generated Products	All non-imagery graphic products, includes map background and route/area overlay.	2	3.7	5
c.4) Radar Product (Single Station)	Includes map background and route/area overlay	2	3.7	5
c.5) National Radar Mosaic	Includes background map and route/area overlay	2	3.7	5
c.6) Satellite Imagery	Includes background map and route/area overlay	2	3.7	5
c.7) Graphic Sequence	4 product single station radar sequence with map background	2	3.7	5
d) Flight Data Retrieval and Display	From data request entry on the user workstation to beginning of display			
d.1) Pre-stored Flight Plan	Single pre-stored flight plan	1.5	3	4
d.2) Preferred Route	Single preferred route	1.5	3	4
d.3) Flight Plan	Single flight plan by aircraft ID	2	3.7	5

Requirement Description	Condition(s) (Product Sizes May Exceed Specified Values)	Time (seconds)		
		Mean	95 th %-tile	99 th %-tile
d.4) Flight Data Lists	Single aircraft ID from Proposed, Inbound, In-flight Contacts, Suspense and Law Enforcement Lists	1	1.8	2.5
d.5) Flight Data List	In-flight contacts	1	1.8	2.5
d.6) ATCSCC Traffic Messages	Any text message of at least 500 bytes	2	3.7	5
d.7) Static flight data	Any static flight data set	3	5.6	7.5
d.8) Application Logon	From Username/Password OK to Functional User Selection	5	9.3	12.5
e) Convert Flight Plan Data	From data request entry on the user workstation to completion of conversion			
e.1) Convert Domestic Flight Plan to ICAO Format	Single domestic flight plan	1	1.8	2.5
e.2) Convert ICAO Flight Plan to Domestic Format	Single ICAO flight plan	1	1.8	2.5
f) Flight Information Updates	From receipt of last byte of update information to completion of DB update.			
f.1) Automatically update flight plan information	ETD change, information received from internal or external source	2	3.7	5
f.2) Automatically update a multiple in-flight work list queue	Information received from internal or external source	2	3.7	5
g) Encode-Decode	From data request entered on the user workstation to response display			
g.1) Encode Request	Single location name entry	2	3.7	5
g.2) Decode Request	Single location identifier entry	2	3.7	5
h) Wild Card Search	From data request entered on the user workstation to response display			
h.1) Encode Request	Single wild card location name (actual first three letters of name provided, state identification not included)	3	5.6	7.5

Requirement Description	Condition(s) (Product Sizes May Exceed Specified Values)	Time (seconds)		
		Mean	95 th %-tile	99 th %-tile
h.2) Decode Request	Single wild card location identifier (2nd or 3rd character wild)	3	5.6	7.5
h.3) History File Transaction Search (last 24 hours)	Single wild card pilot name (first four letters of pilot name)	N/A	N/A	15 min
i) Message Dissemination Within FSS Site	From transmit entry on the user workstation or equivalent to last byte received at appropriate function/position(s)			
i.1) All types of messages	500 bytes, includes flight, weather, alert, and administrative messages	3	5.6	7.5
j) Message Receipt from External Sources	From receipt of last byte of data to update of Proposed, Inbound and Suspense lists. For non-flight plan message updates of alert queue.			
j.1) Non-Flight Plan Messages	500 byte message	1	1.8	2.5
j.2) Flight Plan Messages from Other Alaska Flight Service Facilities	500 byte message	1	1.8	2.5
j.3) Flight Plan Messages from non- Alaska Flight Service Facilities	250 byte message (including flight plan validation)	1	1.8	2.5
k) Message Dissemination to External Sources, except Remote Workstations	From transmit entry on the user workstation or equivalent to last byte received (equivalent means automated initiation of action)			
k.1) Non-flight Plan Messages	500 byte message	0.5	0.9	1.2
k.2) Flight Plan Message	250 byte message (including flight plan validation and automatic addressing)	0.5	0.9	1.2
l) Message Receipt at Remote Workstation	From receipt of last byte to delivery of first byte at appropriate workstation function/position			
l.1) All Data Received from Alaska Flight Service Facility	500 byte message	N/A	N/A	N/A

Requirement Description	Condition(s) (Product Sizes May Exceed Specified Values)	Time (seconds)		
		Mean	95 th %-tile	99 th %-tile
m) Message Dissemination from Remote Workstation	From transmit entry on the Remote Workstation or equivalent to first byte received at Remote Workstation			
m.1) Non-flight Plan Message	500 byte message	0.2	0.4	0.5
m.2) Flight Plan Message	250 byte message	0.2	0.4	0.5
n) Alert Notification Processing	Receipt of last byte of alert message or event parameter met until alert at designated function/position(s)			
n.1) All Alert Notifications	Alert displayed at designated position(s)	2.5	4.6	6.2
o) Flight Data Request	Receipt of request message from user workstation to completion of response message/event report			
o.1) SAR Request	A single aircraft ID, for last 24 hours, transmit response	N/A	N/A	N/A
o.2) Event Reconstruction Request	A single aircraft ID for last 15 days, assemble hardcopy listing	N/A	N/A	30 min
p) Products Display	From data request entry on the user workstation to beginning of display			
p.1) Surface Analysis Plot	Plot	3	5.6	7.5
p.2) Weather Depiction Plot	Plot	3	5.6	7.5
p.3) Radar Plot	Plot	3	5.6	7.5
p.4) Pilot Report Plot	Alaska Flight Service area of responsibility, at least 30 PIREPs	6	11	19
p.5) Severe Weather Forecast Alert Plot	Individual product overlay on map	6	11	19
p.6) SIGMET Plot	Individual product overlay on map	6	11	19
p.7) Convective SIGMET Plot	Individual product overlay on map	6	11	19
p.8) AIRMET Plot	Individual product overlay on map	6	11	19

Requirement Description	Condition(s) (Product Sizes May Exceed Specified Values)	Time (seconds)		
		Mean	95 th %-tile	99 th %-tile
p.9) Center Weather Advisory Plot	Individual product overlay on map	6	11	19
p.10) Special Use Airspace (SUA) Plot	Individual product overlay on map	N/A	N/A	N/A
q) Requested Product	From data request entry on the user workstation to beginning of product display			
q.1) Trend Weather Product	Single weather location	2	3.7	5
q.2) Flight Profile Product	500nm route, 2 legs, corridor width 100nm, low altitude	10	17	27
r) Manually Generated Product	From transmit entry on the user workstation to receipt of first byte			
r.1) Surface Weather Observation	Single observation	0.5	0.9	1.2
r.2) PIREP	Single report	0.5	0.9	1.2
r.3) Data Request (Request-Reply)	Single specialist requests (excludes database restoration requests)	0.5	0.9	1.2
r.4) Data Request (Request-Reply)	Supervisory database restoration request	0.5	0.9	1.2
s) Adjust Viewing Window (Product on Screen)	From scroll or zoom/pan request entry on the user workstation to beginning of action			
s.1) Scroll	Text product	N/A	N/A	N/A
s.2) Zoom	Graphic and image product	N/A	N/A	N/A
s.3) Pan	Continuous to limits of graphic	N/A	N/A	N/A
t) Multi-product Overlays	From request entry on the user workstation to beginning of display of composite product			
t.1) Area radar display and GOES visual imagery	With map background	3	5.6	7.5

Requirement Description	Condition(s) (Product Sizes May Exceed Specified Values)	Time (seconds)		
		Mean	95 th %-tile	99 th %-tile
t.2) Area radar display and GOES visual imagery	With map background and selected flight route	3	5.6	7.5
t.3) SUA and Weather Depiction Plot	With map background and selected flight route	3	5.6	7.5

3.3.2 Performance Management

The Respondent must [AAC-0602] establish and maintain a performance management system to ensure that the appropriate metrics are in place to manage, monitor, and report service performance.

The Respondent must [AAC-0603] maintain the performance management system.

3.3.3 Recovery Requirements

3.3.3.1 Recovery From Power Reset of the System

The AAC system must [AAC-3308] recover to full service after restoration of electrical power to the normal operational mode.

3.3.3.2 Recovery from a System Reset

The AAC system must [AAC-3310] recover to full service from a full system reset defined as a system restart, without the loss of power.

3.3.3.3 Recovery of Essential Data

After a failure or system reset, the AAC system must [AAC-3312] recover all data on the Proposed List, Inbound List, and Suspense List.

After a failure or system reset, the AAC system must [AAC-3313] recover weather and aeronautical databases, the flight plan database, and historically recorded database.

After a data communication loss, the AAC system must [AAC-3314] recover A/N weather data, graphical weather data, aeronautical data, and NOTAM data.

The AAC system must [AAC-3315] provide a method to recover all databases should the databases be corrupted by operations or application fault.

The AAC system must [AAC-3316] recover from a storage device failure.

3.4 Reliability, Maintainability, Availability, and Operational Continuity

Note: An outage is defined as an interruption in the provision of one or more of the primary flight services: preflight, inflight, flight data, and self-assisted services.

3.4.1 Reliability

The Respondent must [AAC-0160] provide service such that in any given calendar month, the total number of service outages does not exceed a total of eight (8) instances. (Critical Performance Requirement)

3.4.2 Maintainability

The Respondent must [AAC-0162] perform maintenance activities without affecting service availability.

3.4.3 Availability

The Respondent must [AAC-1156] measure the duration (in seconds) of each AFS outage attributed to the system lasting 60 seconds or more.

The Respondent must [AAC-1154] provide flight services in Alaska such that operational availability (A_{op}) is greater than or equal to ($\geq$) 0.999, measured monthly. (Critical Performance Requirement)

$$A_{op} = \left[1 - \left(\frac{Total\ Outage\ Time}{Total\ Time} \right) \right] * 100$$

Note: Failures of FAA-provided resources are not required to be reported and will not count against availability.

3.4.4 Continuity of Operations

Continuity of Operations is the set of procedures and actions taken to ensure that flight services are sustained during a Disruptive Incident lasting greater than 12 hours.

A Disruptive Incident is defined as the interruption in the provision or loss of service of any of the primary flight services for more than 120 minutes. Disruptive Incidents may include, but are not necessarily limited to, the following events:

- Force majeure events, such as earthquake, flood, hurricane, terrorist acts, war, or public health outbreak.
- Respondent-induced events, such as system and service outages; Respondent employee work stoppages, slowdowns, or strikes; or human error.

Only Disruptive Incidents due to Respondent-induced events are subject to RMA requirements.

The Respondent must [AAC-0506] provide notifications in the event of an interruption in the provision of any of the primary flight services.

The Respondent must [AAC-0504] develop a Contingency Plan in accordance with FAA Order JO 1900.47, Air Traffic Control Operational Contingency Plans, that describes the plans and procedures to maintain continuity of flight service operations in the event of a disruptive incident.

The Respondent must [AAC-0505] execute the Contingency Plan annually.

The Respondent must [AAC-0507] update the Contingency Plan annually.

4 Integration Requirements

The requirements in Sections 4.1, 4.2, & 4.3 apply to facilities of the system that are owned by the FAA, equipment placed within an FAA facility or operated and maintained by FAA personnel.

4.1 Physical Integration

4.1.1 Environmental Requirements

N/A

4.1.2 Land and Facility

N/A

4.1.3 Telecommunications

N/A

4.2 Functional Integration

4.2.1 Interfaces to Other FAA Enterprise Architecture Elements

The Respondent must [AAC-0302] integrate into the NAS without disruption to existing systems.

The Respondent must [AAC-0308] not degrade the operation of any NAS interface.

The Respondent must [AAC-0304] develop Interface Requirements Documents (IRDs) in accordance with FAA-STD-025F, Preparation of Interface Control Documents.

The Respondent must [AAC-0305] develop Interface Control Documents (ICDs) in accordance with FAA-STD-025F, Preparation of Interface Control Documents.

The Respondent must [AAC-0636] prepare Web Service Requirements Documents (WSRDs) in accordance with FAA-STD-070, Preparation of Web Service Requirements Documents.

The Respondent must [AAC-0638] prepare Web Service Description Documents (WSDDs) in accordance with FAA-STD-065A, Preparation of Web Service Description Documents.

The Respondent must [AAC-0640] prepare Java Message Service Description Documents (JMSDDs) in accordance with FAA-STD-073, Preparation of JAVA Message Service Description Documents.

The Respondent must [AAC-0642] develop Web services in accordance with:

- a. FAA-STD-063, Standard Practice, XML Namespaces
- b. FAA-STD-064, Web Service Registration
- c. FAA-STD-066, Web Service Taxonomies

The Respondent must [AAC-0198] support communication with NADIN via standard Aeronautical Fixed Telecommunication Network (AFTN) messages.

4.2.2 External Interfaces

The Respondent must [AAC-0164] use the FAA's NBPS for all data communications between internal FAA NAS systems (networking and computer infrastructure elements) and Respondent infrastructure.

4.2.2.1 Flight Data

The Respondent must [AAC-0167] interface with NADIN in accordance with NAS-IR-43010002, Interface Requirements Document - National Airspace Data Interchange Network (NADIN) Message Switch Network (MSN) to Transmission Control Protocol (TCP)/Internet Protocol (IP)-based User Subsystems.

The Respondent must [AAC-0165] utilize FAA-provided access to NAS systems for IFR flight plan filing.

The Respondent must [AAC-0169] transmit IFR flight plans to the FAA via NADIN.

The Respondent must [AAC-0163] accept VFR flight plans to support SAR activities.

The Respondent must [AAC-0168] migrate to a future system or service when the FAA decommissions (sunsets) the system from which flight data is received.

The Respondent must [AAC-0196] support communication with the Traffic Flow Management System (TFMS) for IFR flight plan filing.

4.2.2.2 Meteorological Information

The Respondent must [AAC-0171] utilize FAA-provided access to NAS systems to obtain meteorological products.

The Respondent must [AAC-0179] provide users with authoritative meteorological information.

The Respondent must [AAC-0170] interface with WMSCR in accordance with NAS-IR-25070002B, Interface Requirements Document - Weather Message Switching Center Replacement (WMSCR) to Transmission Control Protocol (TCP)/Internet Protocol (IP)-based User Subsystems.

The Respondent must [AAC-0172] migrate to a future system or service when the FAA decommissions (sunsets) the system from which that meteorological information is received.

The Respondent must [AAC-1170] integrate with the FAA Weather Camera system (weathercams.faa.gov) in a manner appropriate to support Alaska Flight Service operations.

4.2.2.3 Aeronautical Information

The Respondent must [AAC-0173] utilize FAA-provided access to NAS systems to obtain aeronautical information.

The Respondent must [AAC-0177] provide users with authoritative aeronautical information.

The Respondent must [AAC-0174] interface with the FAA's NOTAM Manager System.

4.2.3 Spectrum Requirements

N/A

4.3 Human Factors

The AAC system's human factors design must [AAC-4202] be in accordance with FAA Standard HF-STD-001B Human Factors Design Standard (HFDS) for Acquisition of Commercial-Off-the-Shelf Subsystems, Non-Developmental Items, and Developmental Systems.

4.3.1 Usability

The AAC system human system integration must [AAC-4204] be in accordance with the HFDS, Section 5.1.2, Design and Evaluation.

4.3.2 Human-System Interfaces

The AAC system's human-to-system interfaces must [AAC-4206] be in accordance with the HFDS Chapter 5.1.4, Interface.

The AAC system display sizing must [AAC-4207] support TBD screen dimensions.

The AAC EFS display sizing must [AAC-4209] support TBD touchscreen dimensions.

4.3.3 Computer-Human Interfaces (CHI)

The AAC system's human-to-computer interfaces must [AAC-4208] be in accordance with the HFDS, Chapter 5.3.1, General.

The AAC system's human-to-computer interfaces must [AAC-4210] be in accordance with the HFDS, Chapter 5.6, Computer-Human Interface.

4.3.4 General Control and Visual Indicator Information

The AAC system controls must [AAC-4234] be in accordance with the HFDS, Section 5.4.1.1, General Control Information.

The AAC system visual indicators must [AAC-4236] be in accordance with the HFDS, Section 5.4.2.1, General Visual Indicator Information.

4.4 Employee Safety and Health

N/A

5 Security and Safety Requirements

The Respondent must [AAC-0200] comply with NIST SP 800-53 security controls for a “High” impact level system, tailored by the ATO for AAC, in accordance with FAA Order 1370.121, FAA Information Security and Privacy Program & Policy and the following sections.

5.1 Facility Security

N/A

5.2 Personnel Security

The Respondent must [AAC-0412] comply with the personnel security requirements specified in FAA Order 1600.72, Contractor and Industrial Security Program.

5.3 Information Systems Security

The Respondent must [AAC-0644] comply with Federal Information Processing Standards (FIPS) 199 as approved by the ATO Information Systems Security (ISS) organization.

The Respondent must [AAC-0646] comply with FIPS 200 as approved by the ATO ISS organization.

The Respondent must [AAC-5002] comply with AAC security controls listed as a High impact system under Federal Information Processing Standards (FIPS)-199, Standards for Security Categorization of Federal Information and Information Systems.

The Respondent must [AAC-5004] implement the High ATO-tailored NIST SP 800-53 security control requirements. The Respondent must [AAC-5006] implement the system controls identified in the AAC Final Information System Security Assessment (ISSA) Appendix C.

5.3.1 Security Management

The Respondent must [AAC-0648] comply with ATO ISS system and services acquisition requirements.

The Respondent must [AAC-0650] use the ATO ISS Acquisition Handbook and Templates.

The Respondent must [AAC-0652] implement a system security plan.

The Respondent must [AAC-0653] update the system security plan annually.

The Respondent must [AAC-0654] manage the information security architecture.

5.3.2 Security Assessments

The Respondent must [AAC-0656] manage risk assessments.

The Respondent must [AAC-0658] manage security assessments and authorizations.

5.3.3 Security-Related System Support

The Respondent must [AAC-0660] manage configurations.

The Respondent must [AAC-0662] implement contingency plans, procedures and mechanisms.

The Respondent must [AAC-0663] verify contingency operations annually by activating contingency plan.

The Respondent must [AAC-0664] support contingency operations.

The Respondent must [AAC-0668] comply with FAA Order JO 1900.47E, Air Traffic Organization Operational Contingency Plan.

5.3.4 Security and Information Access

The Respondent must [AAC-0670] manage access to information.

The Respondent must [AAC-0674] comply with FAA Order JO 1370.96, ATO System Access Control.

The Respondent must [AAC-0676] comply with FAA Order JO 1370.98, ATO Information Technology (IT) Infrastructure Requirements for Non-FAA connectivity.

The Respondent must [AAC-0678] provide inter-facility communication in accordance with FAA Order JO 1370.114, Implementation of FAA Telecommunications Infrastructure Services and Information Security Requirements in the NAS.

The Respondent must [AAC-0680] manage user identification and authentication.

The Respondent must [AAC-0682] manage identities and credentials for authorized devices and users.

5.3.5 System and Information Monitoring

The Respondent must [AAC-0684] manage audit and accountability information and processes.

The Respondent must [AAC-0688] manage security audit information.

The Respondent must [AAC-0690] implement non-repudiation mechanisms.

The Respondent must [AAC-0692] manage incident responses.

The Respondent must [AAC-0694] manage security event information.

The Respondent must [AAC-0696] comply with the FAA Order 1370.101A, ATO Information Security Incident Reporting and Response Policy.

The Respondent must [AAC-0698] comply with FAA Order JO 1370.46, NAS Information Security Incident Detection, Reporting, and Response.

The Respondent must [AAC-0700] comply with DOT Order 1351.37, Departmental Cybersecurity Program and Compendium (DCC).

5.3.6 System and Information Protection

The Respondent must [AAC-0702] implement system and communication protection procedures.

The Respondent must [AAC-0704] protect system functions and information against security threats.

The Respondent must [AAC-0706] connect to non-NAS networks and non-NAS information systems via the NBPS in accordance with FAA Order JO 1370.114, Implementation of FAA Telecommunications Infrastructure Services and Information Security Requirements in the NAS.

The Respondent must [AAC-0708] manage system and information integrity.

The Respondent must [AAC-0714] protect media.

The Respondent must [AAC-0718] comply with FAA Order 1200.22E, External Requests for National Airspace System (NAS) Data.

The Respondent must [AAC-0720] comply with FAA Order 1600.75, Protecting Sensitive Unclassified Information.

5.3.7 Privacy Information Protection

The Respondent must [AAC-0722] protect Personally Identifiable Information (PII).

5.4 System Safety

The Respondent must [AAC-0420] develop and maintain a Safety Risk Management Document (SRMD) in accordance with the FAA Safety Risk Management Guidance for Systems Acquisition (SRMGSA) and the Safety Management System.

The Respondent must [AAC-0422] conduct program safety assessments in accordance with the FAA ATO Safety Management System Manual and SRMGSA.

The Respondent must [AAC-0424] ensure that the following risk and safety applicable standards and orders are followed:

- a. FAA Order JO 1000.37, Air Traffic Organization Safety Management System,
- b. Safety Risk Management Guidance for System Acquisitions (SRMGSA)
- c. FAA Air Traffic Organization Safety Management System Manual.
- d. FAA Order 8000.369, Safety Management System
- e. FAA Order 8040.4, Safety Risk Management Policy

5.4.1 Existing Controls

TBD

5.4.2 Safety Requirements

TBD

6 Quality Management Requirements

6.1 Quality Assurance

The Respondent must [AAC-0742] implement a quality management plan in accordance with ANSI/ISO/ASQC Q9001:2015, Quality Management Systems.

6.2 Configuration Management

6.2.1 Configuration Management Program

The Respondent must [AAC-0762] implement a configuration management program for software, documentation, engineering/commercial and documentation.

6.2.2 Configuration Management Planning

The Respondent must [AAC-0764] develop a Configuration Management Plan (CMP) in accordance with FAA Order 1800.66, Configuration Management Policy.

The Respondent must [AAC-6010] identify, monitor, and track all software, and documentation related problems and document those problems in Problem Trouble Reports (PTRs).

7 Test and Evaluation Requirements

The Respondent must [AAC-0443] conduct a Test and Evaluation (T&E) program in accordance with FAA AMS T&E Policy and Guidance.

7.1 Planning and Test Design Requirements

The Respondent must [AAC-0444] comply with FAA Test Orders and other test guidance as applicable, including:

- a. FAA Order NG 1810.8A – William J. Hughes Technical Center’s Test and Evaluation Policy
- b. FAA Systems Engineering Manual
- c. FAA William J Hughes Technical Center T&E Handbook
- d. V&V Guidelines in the AMS Toolset
- e. FAA AMS T&E Policy and Guidance

The AAC test program must [AAC-7006] conduct Development Testing (DT), Site Acceptance Testing (SAT), Operational Testing (OT), Field Familiarization (FM) testing, and Regression Testing.

The Respondent must [AAC-0784] maintain a Test & Evaluation Master Plan (TEMP) in accordance with the WJHTC T&E Handbook.

The AAC test program must [AAC-7007] develop and maintain an AAC TEMP that covers all OT planning, testing, and reporting activities in accordance with the WJHTC T&E Handbook.

The Respondent must [AAC-0788] submit a Respondent Master Test Plan (RMTP), which describes the Respondent’s approach to conducting T&E activities.

The Respondent must [AAC-7009] develop and maintain a AAC Respondent Master Test Plan (RMTP), consistent with the AAC Test and Evaluation Master Plan (TEMP) that covers all DT planning, testing, and reporting activities.

7.2 Developmental Test (DT) Requirements

The Respondent must [AAC-7010] conduct an AAC Development Test (DT) to verify system specification requirements, functional design, and integration into the NAS.

The Respondent must [AAC-7016] conduct DT regression testing to verify integrated software and hardware changes did not inadvertently result in a problem elsewhere in the system.

7.2.1 Interoperability Testing

The Respondent must [AAC-0790] submit Acceptance Test Procedures (ATP) for Interoperability Testing.

The Respondent must [AAC-0448] conduct an Interoperability Test to verify that the system meets NAS Interoperability criteria.

The Respondent must [AAC-0449] ensure that the Respondent’s solution does not degrade operation of NAS interfaces.

The Respondent must [AAC-0450] demonstrate compliance with appropriate IRDs, ICDs, WSDDs, WSRDs, and JMSDDs as applicable.

The Respondent must [AAC-0460] verify all interfaces prior to connecting to any NAS systems.

7.2.2 Regression Testing

The Respondent must [AAC-0528] conduct regression testing when changes have been made to previously tested system components.

7.2.3 Key Site Testing

The Respondent must [AAC-7031] identify testing to be conducted at AAC field sites and key sites.

7.2.4 Site Acceptance Testing

The Respondent must [AAC-7015] conduct an AAC Site Acceptance Testing (SAT) to verify site installation, system interface, and operational configuration at each installation site.

7.3 Operational Test (OT) Requirements

The Respondent must [AAC-0452] conduct Cutover Procedure Verification to validate the Respondent's cutover procedures.

The Respondent must [AAC-7017] conduct OT at the WJHTC to validate that AAC services/systems are operationally effective and suitable for maintenance and operations in the NAS, and the NAS infrastructure is ready to accept the AAC services/systems.

The AAC OT must [AAC-7018] ensure it has no adverse effect on the rest of the NAS.

The AAC OT must [AAC-7019] evaluate and resolve each AAC Critical Operational Issue (COI), Critical Performance Requirement (CPR), and program requirement being addressed in OT.

The AAC OT Test Team must [AAC-7030] conduct regression testing to validate the integrity of solutions to Discrepancy Reports/Program Trouble Report (DR/PTR) have not introduced new problems or issues and to support validation of the system's readiness for operational use.

7.4 Reporting Requirements

The Respondent must [AAC-0786] document Acceptance Test Reports (ATR) for the following:

- a. Development Testing
- b. SAT
- c. Interoperability Testing

The Respondent must [AAC-0534] ensure Acceptance Test Reports include the tests conducted, the data collected, the data recording and analysis process, the requirement verification and validation status, conclusions and recommendations.

The AAC Development Test reports must [AAC-7035] document the status of the specification requirements, contractual requirements and functional design of the AAC system.

7.5 Test Support Requirements

The Respondent must [AAC-0810] ensure participating test personnel are trained for testing the products under evaluation.

The Respondent must [AAC-0812] ensure test readiness criteria are met prior to the Test Readiness Review.

The Respondent must [AAC-0530] connect to the FTI National Test Bed (FNTB) at the William J. Hughes Technical Center (WJHTC) and/or the FAA Cloud National Test Bed (CNTB) to verify requirements.

The Respondent must [AAC-7040] provide new AAC test assets including test tools, test simulations, test adaptations, test configurations, and test bed environments prior to test conduct when they are needed to support requirements evaluation, maintenance, and certification.

The Respondent must [AAC-0532] conduct Interoperability testing at the Respondent designated facilities.

The Respondent must [AAC-7039] utilize test tools that emulate NAS interfaces during development when the actual test interface is not available.

7.6 Critical Operational Issues

The following operational issues have been identified by the AAC program office as Critical Operational Issues (COIs) for the provision and execution of AAC.

COI-A1: Do the AAC services provide suitable and effective support to Flight Service Specialist in performing pilot weather briefings in accordance with FAA Order 7110.10?

COI-A2: Do the AAC services provide suitable, effective, and timely support to Flight Service Specialist in performing all flight planning functions in accordance with FAA Order 7110.10?

COI-A3: Are the AAC services capable of receiving, recognizing, pre-processing, storing, and transmitting weather, flight plan, search and rescue to and from multiple external sources?

COI-A4: Do the AAC systems interface and operate with NAS interfaces for data exchange without degradation?

COI-A5: Can the AAC AAC systems be maintained according to the maintenance philosophy (including certification and remote maintenance), without disruption or degradation to operations?

COI-A6: Does the AAC solution have the reliability, maintainability, and availability to provide for flight services in accordance with the requirements of the contract?

COI-A9: Does the AAC solution accommodate the expected traffic load without significant degradation of service under load?

COI-A10: Does the AAC provide adequate security controls and mitigation of security risks for the solution?

COI-A13: Does the AAC support the transfer of AOR between FSSs?

8 In-Service Support and Management

8.1 Integrated Logistics Support

TBD

8.2 Maintenance Planning

TBD

8.3 Packaging, Handling, Storage, and Transportation

N/A

8.4 Support Equipment Maintenance

8.4.1 Tools and Test Equipment

N/A

8.4.2 Maintenance Operations

The Respondent must [AAC-0848] provide advanced notification 48 hours prior to performing activities that have the potential to impact Air Traffic service, other FAA systems or facilities, or the provision of service to flight service users.

The Respondent must [AAC-0850] coordinate with Technical Operations in accordance with FAA Order 6000.15, General Maintenance Handbook for Airways Facilities, for outages related to government provided services.

8.5 Training, Training Support, and Personnel Skills

The Respondent must [AAC-0842] implement a training program to prepare operational personnel to meet and maintain certification requirements.

The Respondent must [AAC-0844] describe its approach to training operational personnel in a Training Program Plan (TPP).

The Respondent must [AAC-0846] develop a training program in alignment with:

- a. FAA Order JO 3120.4 Air Traffic Technical Training
- b. FAA Order JO 7220.4 FAA Certification of Pilot Weather Briefing.

8.5.1 FAA Academy Training System

N/A

8.5.2 AAC Training, Training Support, and Personnel Skills

The Respondent must [AAC-9058] develop and deliver AAC training in accordance with FAA Order 3000.22B using FAA-STD-028C as supplemental guidance.

8.5.2.1 Technical Operations Training

TBD

8.5.2.2 Air Traffic Training

The AAC will have the Alaska Flight Service Training Academy (AFSTA) Cadre of Core trainers provide initial and recurring training of Certified Professional Controllers (CPC), Front Line Managers, Operations Managers, Staff Specialists, Air Traffic Managers

The Respondent must [AAC-9064] ensure System Operations Air Traffic (AT) training is in accordance with FAA Order JO 3120.4R, Air Traffic Technical Training with emphasis on Appendix C, Flight Service Instructional Program Guide.

The Respondent must [AAC-9066] provide FAA Cadre of Core Trainers initial and recurring AAC training.

8.5.2.2.1 CPC Training Support Processing

The Respondent must [AAC-9068] provide an on-line tutorial for training support processing.

8.5.2.2.2 Configuration Training Mode

The Respondent will provide a configuration training mode that allows the creation and storage of scripted scenarios as well as the capture of live weather data to use for operational training.

The Respondent must [AAC-1993] provide a configuration training mode.

The Respondent must [AAC-1994] enter the configuration training mode without operational impact.

8.5.2.2.3 Scenarios

In configuration training mode, the system will capture and store live weather and aeronautical data. The system will also allow the creation, storage, retrieval, edit and deletion of scripted training scenarios containing stored weather data and other messages.

The Respondent must [AAC-1995] capture live alphanumeric weather data.

The Respondent must [AAC-1996] capture live aeronautical data.

The Respondent must [AAC-1997] capture live graphical weather data.

The Respondent must [AAC-1998] store live alphanumeric weather data.

The Respondent must [AAC-1999] store live aeronautical data.

The Respondent must [AAC-2000] store live graphical weather data.

The Respondent must [AAC-2001] provide the capability to create scripted scenarios.

The Respondent must [AAC-2002] provide the capability for scripted scenarios to contain stored alphanumeric weather data.

The Respondent must [AAC-2003] provide the capability for scripted scenarios to contain stored aeronautical data.

The Respondent must [AAC-2004] provide the capability for scripted scenarios to contain stored graphical weather data.

The Respondent must [AAC-2005] provide the capability for scripted scenarios to contain messages.

The Respondent must [AAC-2006] provide the capability to store scripted scenarios.

The Respondent must [AAC-2007] provide the capability to retrieve scripted scenarios.

The Respondent must [AAC-2008] provide the capability to edit scripted scenarios.

The Respondent must [AAC-2009] provide the capability to delete scripted scenarios.

8.5.2.2.4 Operational Training Mode

The automation system will provide an operational training mode that allows the replay of stored scenarios in scripted mode as well as messages generated and injected by an instructor in interactive mode.

The Respondent must [AAC-2010] provide an operational training mode.

The Respondent must [AAC-2011] enter the operational training mode without operational impact.

The Respondent must [AAC-2012] provide a scripted operational training mode.

The Respondent scripted mode must [AAC-2013] provide the capability to send data in a timed sequence.

The Respondent must [AAC-2014] provide an interactive operational training mode.

The Respondent interactive mode must [AAC-2015] provide an instructor with the capability to interactively generate messages.

The Respondent interactive mode must [AAC-2016] provide an instructor with the capability to interactively send messages.

8.5.2.2.5 Scenario Replay

Scenarios are scripts containing stored alphanumeric weather data, weather graphics and other messages. Scenarios will be stored on the System and can be replayed to simulate conditions for training.

The Respondent must [AAC-2017] provide the capability to replay scripted scenarios on any workstation without operational impact.

The Respondent must [AAC-2018] provide the capability to start multiple scripts during a single session replay.

The Respondent must [AAC-2019] provide the capability to start new scripts once a session replay has already begun.

The Respondent must [AAC-2020] provide a “start” command within the scenario to begin replay. The Respondent must [AAC-2021] provide a “stop” command within the scenario during replay.

The Respondent must [AAC-2022] provide a “resume” command within the scenario during replay.

The Respondent must [AAC-2023] provide an instantaneous “fast forward” command to specific times within the scenario during replay.

The Respondent must [AAC-2024] provide an instantaneous “rewind” command to specific times within the scenario during replay.

The Respondent must [AAC-2025] record each transaction entry during replay.

8.5.2.2.6 Import of Weather Data

The Respondent must [AAC-2026] provide the capability to import the stored alphanumeric weather data.

The Respondent must [AAC-2027] provide the capability to import the stored aeronautical data.

The Respondent must [AAC-2028] provide the capability to import the stored graphical weather data.

8.5.2.2.7 Export of Weather Data

The Respondent must [AAC-2029] provide the capability to export the stored alphanumeric weather.

The Respondent must [AAC-2030] provide the capability to export the stored aeronautical data.

The Respondent must [AAC-2031] provide the capability to export the stored graphical weather data.

8.5.2.3 FAA System Administrators and Security Administrators Training

The Respondent must [AAC-9100] ensure the AAC System Administrators and Security Administrators training includes management of system access such as user privileges, password resets, etc.

8.6 Technical Data

TBD

Appendix A Critical Performance Requirements

Critical performance requirements are primary requirements that represent attributes or characteristics considered essential to meeting the mission need the program is seeking to satisfy. Critical performance requirements are a part of total program requirements that define the operational framework and performance baseline for the investment program.

Table 5 - Critical Performance Requirements

CPR #	PRD #	PRD Requirement Text	Quantitative/Qualitative Value Type
CPR-06	AAC-0160	The Respondent must provide Alaska flight services such that in any given calendar month, the total number of service outages does not exceed a total of eight (8) instances.	Quantitative
CPR-07	AAC-0154	The Respondent must provide Alaska flight services such that A_{op} is greater than or equal to ($\geq$) 0.999, measured on a monthly basis.	Quantitative

Appendix B Acronyms and Glossary

B.1 Acronyms and Abbreviations

The following table contains the acronyms and abbreviations used within this document.

Table 6: Acronyms and Abbreviations

Acronym	Description
AAC	Alaska Automation Capability
ADIZ	Air Defense Identification Zone
ADS-B	Automatic Dependent Surveillance – Broadcast
AES	Advanced Encryption Standard
AFS	Alaska Flight Service
AFSS	Automated Flight Service Station
AFSTA	Alaska Flight Service Training Academy
AFTN	Aeronautical Fixed Telecommunication Network
A/G	Air to Ground
AIRMET	Airman’s Meteorological information
AISR	Aeronautical Information System Replacement
AJM	FAA Program Management Organization
AJW	FAA Technical Operations Organization
AMOC	Air Marine Operations Center
AMS	Acquisition Management System
ANG	FAA Next Gen Organization
ANSI	American National Standards Institute
A _{op}	Operational Availability
AOR	Area of Responsibility
ARTCC	Air Route Traffic Control Center

Acronym	Description
ASQC	American Society for Quality Control
ASTI	Alaska Satellite Telecommunications Infrastructure
ASTM	American Society for Testing and Materials
ATC	Air Traffic Control
ATCRBS	Air Traffic Control Radar Beacon System
ATCSCC	Air Traffic Control System Command Center
ATO	Air Traffic Organization
ATP	Acceptance Test Procedures
ATR	Acceptance Test Report
ATSS	Airway Transportation System Specialist
CBP	U.S. Customs and Border Protection
CFR	Code of Federal Regulations
CHI	Computer Human Interface
CIC	Controller In Charge
CMP	Configuration Management Plan
COI	Critical Operational Issue
CONUS	Continental United States
CNTB	Cloud National Test Bed
CPC	Certified Professional Controller
CPR	Critical Performance Requirement
CSS	Common Support Services
CWP	Controller Working Position
DCC	Departmental Cybersecurity Program and Compendium
DOS	Description of Services
DOT	Department of Transportation

Acronym	Description
DR	Discrepancy Report
EFS	Electronic Flight Strip
EOSH	Environmental and Occupational Safety and Health
eSRS	Enhanced Special Reporting Service
EUROCAE	European Organization for Civil Aviation Equipment
FAA	Federal Aviation Administration
FAR	Federal Aviation Regulation
FAT	Factory Acceptance Testing
FENS	FAA Enterprise Network Services
FDC	Flight Data Center
FFSP	Future Flight Service Program
FIPS	Federal Information Processing Standard
FNTB	FTI National Test Bed
FPA	Flight Plan Area
FRZ	Flight Restriction Zone
FSNESS	Flight Service National Airspace System Efficient Streamlined Services
FSP	Facility Security Plan
FTI	FAA Telecommunications Infrastructure
GA	General Aviation
GFE	Government Furnished Equipment
GOES	Geostationary Operational Environmental Satellite
GPS	Global Positioning System
HFDS	Human Factors Design Standard
hr	Hour
HVAC	Heating, Ventilation, and Air Conditioning

Acronym	Description
ICAO	International Civil Aviation Organization
ICD	Interface Control Document
IFR	Instrument Flight Rule
ILS	Integrated Logistics Support
ILSP	Integrated Logistics Support Plan
IP	Internet Protocol
IRD	Interface Requirements Document
ISM	In-Service Management
ISO	International Standards Organization
ISPD	Implementation Strategy and Planning Document
ISS	Information Systems Security
ISSA	Information Systems Security Analysis
IT	Information Technology
LOCID	Location Identification
ms / msec	millisecond (= 0.001 second)
MSN	Message Switch Network
MSR	Monthly Status Report
N/A	Not Applicable
NADIN	National Airspace Data Interchange Network
NAS	National Airspace System
NBPS	NAS Boundary Protection System
NMS	NOTAM Management System
NIST	National Institute of Standards and Technology
NOAA	National Oceanic and Atmospheric Administration
NOTAM	Notice to Airmen

Acronym	Description
O&SHA	Operating & Support Hazard Analysis
OASIS	Operational and Supportability Implementation System
OJT	On-the-Job Training
OPM	Operational Procedure Manuals
ORD	Operational Readiness Decision
OSD	Operational Suitability Demonstration
OSHA	Occupational Safety and Health Administration
PCA	Physical Configuration Audit
PHA	Preliminary Hazard Assessment
PII	Personally Identifiable Information
PLA	Program/Project Level Agreement
PMP	Program Management Plan
PSP	Program Safety Plan
PSTN	Public Switched Telecommunications Network
PTR	Program Trouble Report
PWS	Performance Work Statement
QA	Quality Assurance
QRO	Quality Reliability Officer
RCM	Reliability Centered Maintenance
RID	
RMA	Reliability, Maintainability and Availability
RMTP	Respondent Master Test Plan
SAR	Search and Rescue
SAT	Site Acceptance Test
SCN	Specification Change Notice

Acronym	Description
SFRA	Special Flight Rules Area
SIGMET	Significant Meteorological Information
SHA	Systems Hazard Analysis
SIR	Screening Information Request
SLE	Second Level Engineering
SMP	Security Management Plan
SMS	Safety Management System
SRM	Safety Risk Management
SRMD	Safety Risk Management Document
SRMGSA	Safety Risk Management Guidance for Systems Acquisition
SSC	System Support Center
SSAR	System Safety Assessment Report
SSMP	System Safety Management Program
STMP	Special Traffic Management Program
STVS	Small Tower Voice System
SUA	Special Use Airspace
SVFR	Special Visual Flight Rules
SWIM	System Wide Information Management
TCP	Transfer Control Protocol
TDM	Time-Division Multiplexing
TEMP	Test & Evaluation Master Plan
TFMS	Traffic Flow Management System
TFR	Temporary Flight Restriction
TIC	Trusted Internet Connection
TP	Transition Plan

Acronym	Description
TPP	Training Program Plan
TRR	Test Readiness Review
USNOF	United States NOTAM Office
USSS	United States Secret Service
UTC	Universal Coordinated Time
VCS	Voice Communication System
VFR	Visual Flight Rules
VHF	Very High Frequency
VIP	Very Important Person
VoIP	Voice over Internet Protocol
VOR	VHF Omnidirectional Radio Range
VOX	Voice Activated Circuitry
VRS	Voice Recording System
VRTM	Verification Requirements Traceability Matrix
WJHTC	William J. Hughes Technical Center
WMSCR	Weather Message Switching Center Replacement
WSR	Weather Surveillance Radar
WSRDD	Web Services Requirements Description Document
XML	eXtensible Markup Language

B.2 Terms and Definitions

Important terms used in this document or relevant to the content are defined in the table below.

Table 7: Terms and Definitions

Term	Definition
Aircraft Orientation	Assist the pilot in determining aircraft's position in emergency and non-emergency situations.
Air Defense Identification Zone (ADIZ)	An area of airspace over land or water (border and offshore coastal areas) in which the ready identification, location, and control of civil aircraft is required in the interest of national security
Authenticated User	An external user (e.g. pilot) of the system, who has previously provided appropriate authorizing credentials for access and use of the system services. Authentication is a process in which the credentials provided are compared to those on file in a database of authorized users' information. If the credentials match, the process is completed and the user is granted authorization for access.
Availability	The probability that a system or constituent piece will be operational during any randomly selected period of time, or, alternatively, the fraction of the total available operating time that the system or constituent piece is operational.
Broadcast Services	A service which disseminates meteorological and aeronautical information to pilots through live transmissions, pre-recorded messages, and voice-to-text messages. This service requires knowledge of current and forecast meteorological information, the ability to rapidly transmit information regarding changes to the National Airspace System (NAS), and response capabilities to special requests for assistance from other Air Traffic Control (ATC) facilities.
Continuity of Operations	The set of procedures and actions taken to ensure that flight services will be sustained during a Disruptive Incident lasting greater than 12 hours.
Defense Visual Flight Rules	Rules applicable to flights within an ADIZ conducted under the visual flight rules in FAR Part 91.
Discrete Beacon Code / Discrete Transponder Code	As used in the Air Traffic Control Radar Beacon System (ATCRBS), any one of the 4096 selectable Mode 3/A aircraft transponder codes except those ending in "zero zero," e.g., discrete codes: 0010, 1201, 2317, 7777; non-discrete codes: 0100, 1200, 7700. Non-discrete codes are normally reserved for radar facilities that are not equipped with discrete decoding capability and for other purposes such as emergencies (7700), VFR aircraft (1200), for example. See FAA Order JO 7110.66, National Beacon Code Allocation Plan

Term	Definition
Disruptive Incident	A Disruptive Incident is defined as the interruption in the provision or loss of service of any of the primary flight services for more than 120 minutes.
Enhanced Air Defense Identification Zone (ADIZ) Coordination	Notify pilots of prohibited/restricted area of domestic airspace where the ready identification, location, and control of aircraft are required in the interests of national security.
Emergency Services	Provide services to aircraft in emergency or distress situations. This priority service requires the specialist to quickly provide comprehensive information to pilots and emergency personnel.
FAA Approved Web Portal or link	Internet based access to flight service information compliant with Information Services and Information System Security orders, notices, policies, regulations and/or industry/ FAA standards.
Flight Data Center Notices to Airmen (FDC NOTAM)	A NOTAM that refers to information that is normally regulatory in nature, transmitted by USNOF and given system wide dissemination.
Flight Data Services	Provide data processing/editing services for aircraft flight information, as well as the dissemination of operational, aeronautical and meteorological message traffic.
Flight Plan	Specified information relating to the intended flight of an aircraft that is filed orally or in writing with a Flight Service Station or an Air Traffic Control facility.
Flight Plan Services	Accept, validate, and process pilot flight plan proposals. Provide a customized service tailored to the needs and knowledge of the pilot. Requires the use of current aeronautical charts, the latest edition of pertinent flight supplements, Notices to Airmen (NOTAMs), and International NOTAMs, as well as knowledge of local weather phenomenon, topography, and alternative route information.
Flight Restricted Zone (FRZ) Flight Plans	Flight plans filed by pilots intending to operate within a 15-mile radius of Washington, D.C. There are numerous security restrictions associated with the operation of FRZ flight plans
General Aviation (GA)	That portion of civil aviation which encompasses all facets of aviation except air carriers holding a certificate of public convenience and necessity from the Civil Aeronautics Board and large aircraft commercial operators.
Inbound List	A list of active flights within the area of responsibility of a given FSS.

Term	Definition
Inflight Services	Customized services, tailored to the needs and knowledge of a pilot while airborne. Services include updates due to changing meteorological and aeronautical conditions as well as providing route information concerning restricted flight zones, relaying IFR clearances, providing lost aircraft orientation, for example.
Inflight Services – Routine Radio Contacts	Consist of information received from, or provided to, en route pilots which can include weather conditions, flight plan activation and closure, and other services
Instrument Flight Rules (IFR)	Rules governing the procedures for conducting instrument flight. Also a term used by pilots and controllers to indicate type of flight plan.
International Cautionary Advisory	An advisory issued to a pilot planning a flight outside of United States controlled airspace. Phraseology: “Check data as soon as practical after entering foreign airspace, as our international data may be inaccurate or incomplete.
International Civil Aviation Organization (ICAO)	A specialized agency of the United Nations whose objective is to develop the principles and techniques of international air navigation and to foster planning and development of international civil air transport.
International Operations	Flight movement messages for civil aircraft operating between the United States and foreign locations and flights within foreign countries.
Law Enforcement and Public Safety Support	Support services within the capabilities of the service provider are provided to law enforcement and public safety organizations as requested. Examples include emergency response notifications and assisting in the coordination of disseminating information to other involved agencies/organizations.
National Airspace System (NAS)	The common network of U.S. airspace; air navigation facilities, equipment and services, airports or landing areas; aeronautical charts, information and services; rules, regulations and procedures, technical information, and manpower and material. Included are system components shared jointly with the military.
National Security	Disseminate national security and airspace information through the United States government, as well as information concerning domestic and foreign aviation interests. Provide flight plan and inflight services to ensure that pilots avoid restricted areas and to facilitate the orderly and controlled movement of authorized aircraft through sensitive areas.
Notice To Airmen (NOTAM)	A notice containing information (not known sufficiently in advance to publicize by other means) concerning the establishment, condition, or change in any component (facility, service, or procedure of, or hazard in the National Airspace System) the timely knowledge of which is essential to personnel concerned with flight operations.

Term	Definition
Notices to Airmen (NOTAM) Entry and Dissemination	NOTAMs contain information about conditions within the National Airspace System that could affect flight operations. Airport owners and operators as well as FAA personnel issue this data by relaying it to Flight Service personnel, who then formats the NOTAM for widespread distribution. Some airport and FAA personnel currently enter NOTAMs directly into the NOTAMs distribution system using a web-based NOTAM entry system.
Operator	An operator of the system (e.g. Flight Service Specialist, maintenance user) who has direct authorized access to the system.
Outage	The inability of the service provider to provide one or more of the primary flight services: preflight, inflight, flight data, and self-assisted services.
Pilot	An external user of the system (who may or may not be a licensed pilot) but who makes use of the flight services provided. Also referred to as a “User” and contrasting with “Specialist” or “Operator.”
Pilot Briefings	A service provided by the Flight Service Specialist to assist pilots in flight planning. Briefing items may include meteorological information, NOTAMs, military activities, flow control information, and other items as requested.
Pilot Weather Reports (PIREPs)	Reports of meteorological phenomena encountered by aircraft in flight.
Position Report	A report over a known location as transmitted by an aircraft to ATC.
Primary Flight Services	Primary flight services include preflight, inflight, flight data, and self-assisted services.
Proposed List	A list of VFR flight plans that have not been activated and IFR flight plans that have not been transmitted to ARTCC.
Round-Robin Flight	A flight where the origin and final destination are the same, but which includes at least one additional intermediate destination.
Search and Rescue (SAR)	A service which seeks missing aircraft and assists those found to be in need of assistance. It is a cooperative effort using the facilities and services of available Federal, state and local agencies. The U.S. Coast Guard is responsible for coordination of search and rescue for the Maritime Region, and the U.S. Air Force is responsible for search and rescue for the Inland Region. Information pertinent to search and rescue should be passed through any air traffic facility or be transmitted directly to the Rescue Coordination Center by telephone.
Self-Assisted User Actions	All on-line/computer-based actions performed by the user when accessing the system providing self-assisted services.

Term	Definition
Special Flight Rules Area (SFRA)	An area of airspace over land or water (border and offshore coastal areas) in which the ready identification, location, and control of civil aircraft is required in the interest of national security
Special Traffic Management Program (STMP)	Reservations required for aircraft bound to or from special high traffic-volume events, such as major sporting events.
Special Use Airspace (SUA)	Airspace of defined dimensions identified by an area on the surface of the earth wherein activities must be confined because of their nature and/or wherein limitations may be imposed upon aircraft operations that are not a part of those activities.
Special VFR Operations	Aircraft operating in accordance with clearances within Class B, C, D, and E surface areas in weather conditions less than the basic VFR weather minima. Such operations must be requested by the pilot and approved by ATC.
Specialist	A flight services specialist who acts as a system operator, providing certified flight services to users. See also "Operator." Contrasts with "User" or "Pilot" who are external users of the system.
Suspense List	A list of Service-B messages that are awaiting acknowledgment.
Temporary Flight Restriction (TFR)	A TFR is a regulatory action issued by the FAA via the U.S. NOTAM System, under the authority of United States Code, Title 49. TFRs are issued within the sovereign airspace of the United States and its territories to restrict certain aircraft from operating within a defined area on a temporary basis to protect persons or property in the air or on the ground. While not all inclusive, TFRs may be issued for disaster or hazard situations such as: toxic gas leaks or spills, fumes from flammable agents, aircraft accident/incident sites, aviation or ground resources engaged in wildlife suppression, or aircraft relief activities following a disaster. TFRs may also be issued in support of VIP movements; for reasons of national security; or when determined necessary for the management of air traffic in the vicinity of aerial demonstrations or major sporting events. NAS users or other interested parties should contact a Specialist for TFR information. Additionally, TFR information can be found in automated briefings, NOTAM publications, and on the internet at http://www.faa.gov . The FAA also distributes TFR information to aviation user groups for further dissemination.
Tie-in Facility	The tie-in facility is the Flight Service facility primarily responsible for providing flight services for a given flight.
Trans-border Operations (Mexico and Canada)	Flight movement messages transmitted between the United States/Mexican border and United States/Canadian border.

Term	Definition
User	A user of the system or service provided (e.g. pilot or airport manager). Also referred to as “pilot”. Compare/contrast with “Operator” and “Specialist.”
Visual Flight Rules	Rules that govern the procedures for conducting flight under visual conditions. The term “VFR” is also used in the United States to indicate weather conditions that are equal to or greater than minimum VFR requirements. In addition, it is used by pilots and controllers to indicate type of flight plan.
Weather Briefings	Meteorological and aeronautical information provided to pilots, flight crews and aviation personnel. Requires the translation, interpretation, and validation of numerous alphanumeric and graphic weather products into a directly usable and understandable format.

Appendix C Reference Documents

The references cited within this document are listed below.

The specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless specifically identified, the issue of a document that is cited in this document is the one (including all change pages, amendments, inserts, supplemental reports) that is current at the time of contract award.

C.1 FAA Documents

Copies of FAA specifications, standards, and publications not available on the FAA's public website will be provided by the FAA upon receipt of the appropriate disclosure.

- FAA AMS T&E Policy and Guidance
- FAA ATO ISS Acquisition Handbook and Templates
- FAA ATO NAS Modernization System Safety Management Program (SSMP)
- FAA ATO Safety Management System Manual.
- FAA Order NG 1810.8A, William J. Hughes Technical Center's Test and Evaluation Policy
- FAA NAS-IR-25070002B, Interface Requirements Document - Weather Message Switching Center Replacement (WMSCR) to Transmission Control Protocol (TCP)/Internet Protocol (IP)-based User Subsystems
- FAA NAS-IR-43010002, Interface Requirements Document - National Airspace Data Interchange Network (NADIN) Message Switch Network (MSN) to Transmission Control Protocol (TCP)/Internet Protocol (IP)-based User Subsystems.
- FAA Order 1200.22E, External Requests for National Airspace System (NAS) Data
- FAA Order 1370.101A, ATO Information Security Incident Reporting and Response Policy
- FAA Order 1370.121, FAA Information Security and Privacy Program & Policy.
- FAA Order 1600.2, Classified National Security Information
- FAA Order 1600.69, FAA Facility Security Management Program.
- FAA Order 1600.72, Contractor and Industrial Security Program.
- FAA Order 1600.75, Protecting Sensitive Unclassified Information
- FAA Order 1800.66, Configuration Management Policy
- FAA Order 6000.15, General Maintenance Handbook for Airways Facilities
- FAA Order 8000.369, Safety Management System Guidance
- FAA Order 8040.4, Safety Risk Management
- FAA Order JO 1000.37, Air Traffic Organization Safety Management System, Safety Risk Management Guidance for System Acquisitions (SRMGSA)
- FAA Order JO 1370.46, NAS Information Security Incident Detection, Reporting, and Response
- FAA Order JO 1370.96, ATO System Access Control

- FAA Order JO 1370.98, ATO Information Technology (IT) Infrastructure Requirements for Non-FAA connectivity
- FAA Order JO 1370.114, Implementation of FAA Telecommunications Infrastructure Services and Information Security Requirements in the NAS
- FAA Order JO 1900.47, Air Traffic Control Operational Contingency Plans
- FAA Order JO 3120.4 Air Traffic Technical Training
- FAA Order JO 7110.10, Flight Services
- FAA Order JO 7210.3, Facility Operation and Administration
- FAA Order JO 7220.4 FAA Certification of Pilot Weather Briefing.
- FAA Systems Engineering Manual
- FAA V&V Guidelines in the AMS Toolset
- FAA William J Hughes Technical Center Test and Evaluation Handbook
- FAA-STD-025F, Preparation of Interface Documentation
- FAA-STD-063, Standard Practice, XML Namespaces
- FAA-STD-064, Web Service Registration
- FAA-STD-065A, Preparation of Web Service Description Documents
- FAA-STD-066, Web Service Taxonomies
- FAA-STD-070, Preparation of Web Service Requirements Documents
- FAA-STD-073, Preparation of JAVA Message Service Description Documents

C.2 Other Government Documents

Copies of federal publications may be obtained from the U.S. Government Printing Office, 710 North Capitol Street, Washington DC, 20401, by calling (202) 512-0132, or through the web site <http://bookstore.gpo.gov/>.

- ANSI/ISO/ASQC Q9001:2015, Quality Management Systems
- DOT Order 1351.37, Departmental Cybersecurity Program and Compendium (DCC)
- FIPS 199
- FIPS 200
- NIST SP 800-53

C.3 Non-Government Documents

Non-government documents may be requested from the organization responsible for the development and dissemination of the required document.

- ICAO Rules of the Air and Air Traffic Services document 4444